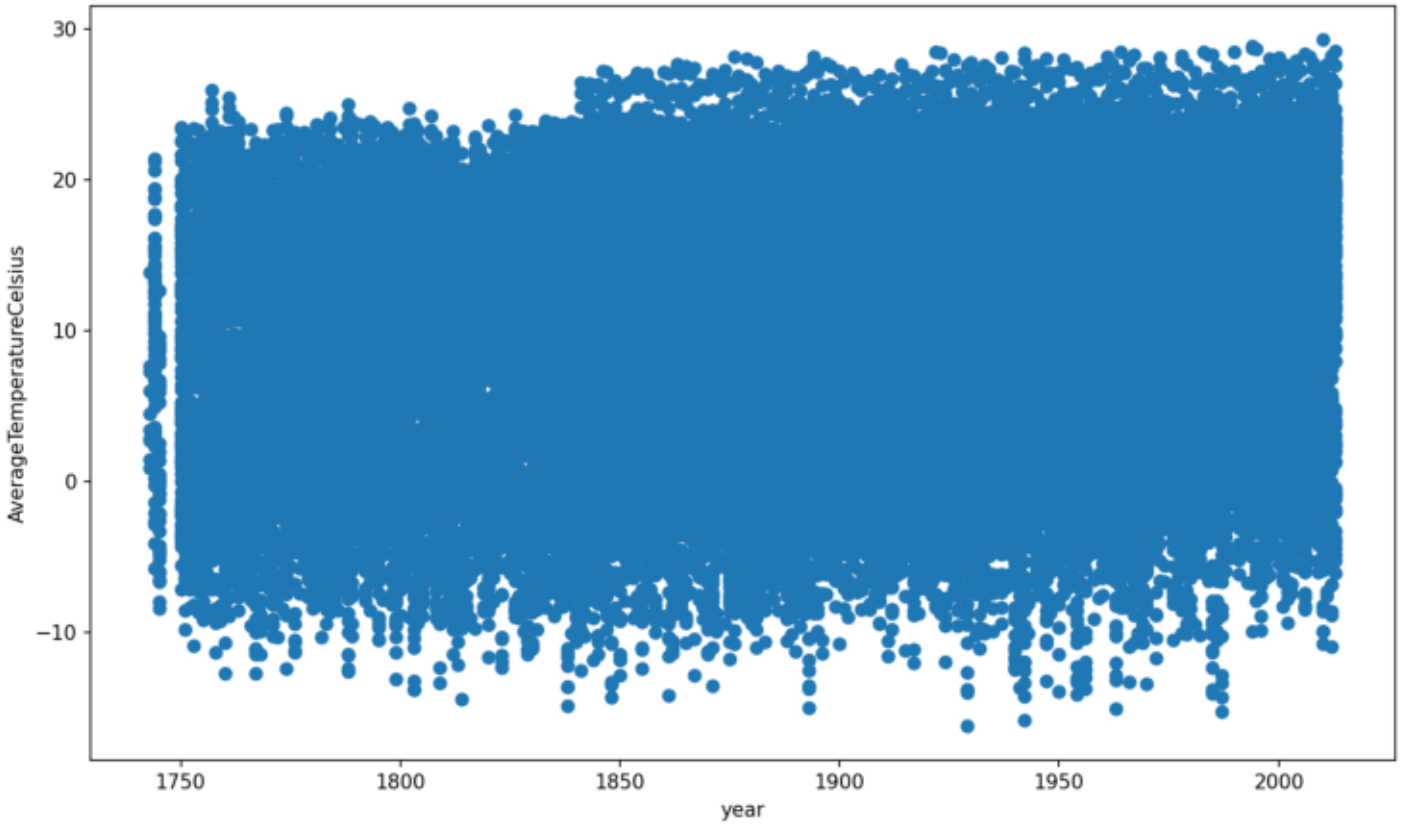
Exploring Global Temperatures Since 1750

Data Analysis & Visualization - Labs 4 & 5 (temperature project)

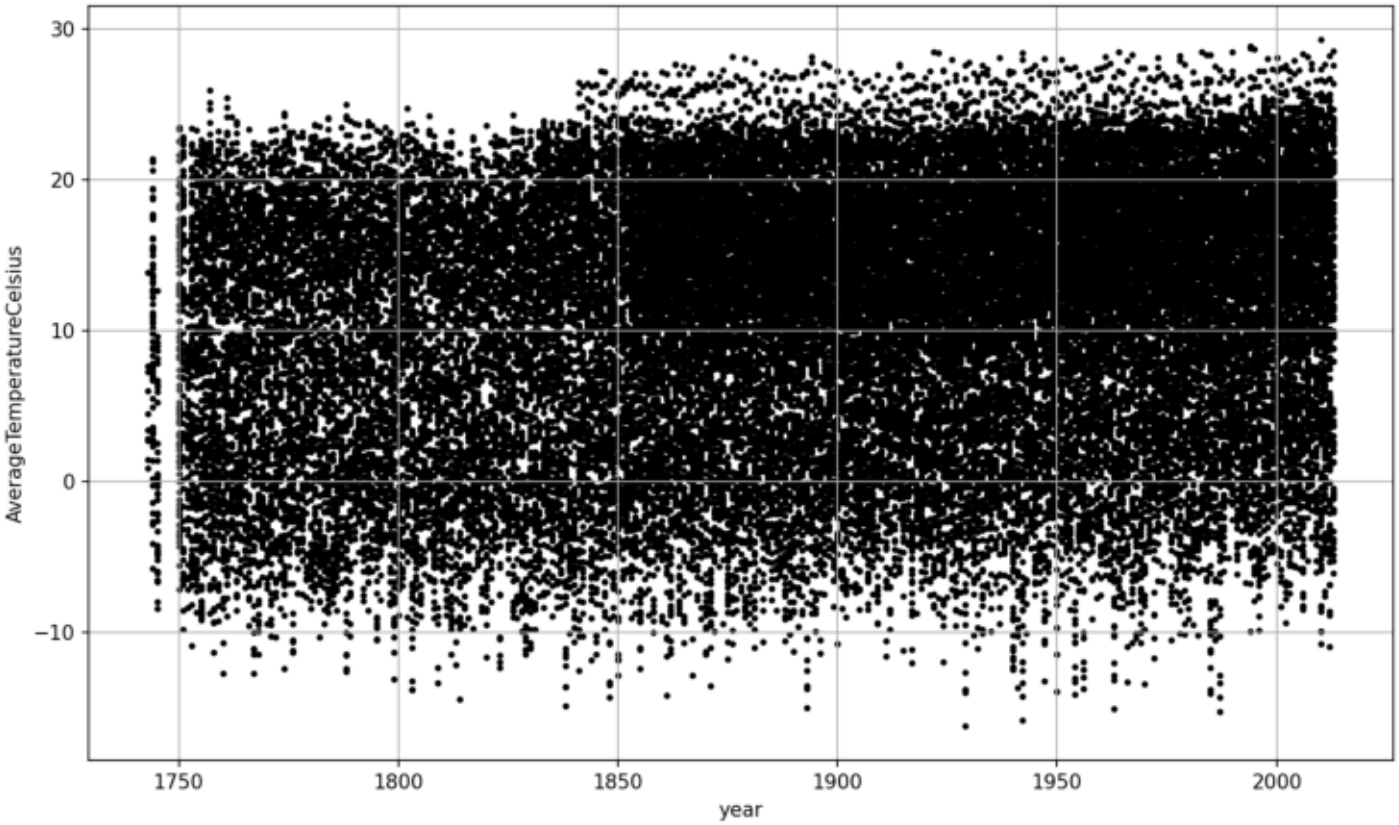
This report walks through the temperature-exploration project from the simplest to progressively more informative plots. Task 1 cleaned the raw data (temperatures_clean.csv): rows with missing City/Country or missing Fahrenheit temperature/uncertainty were dropped, the constant 'day' column was removed, and temperatures were converted from Fahrenheit to Celsius. Tasks 2-3 (lab 4) build scatter, box and violin plots of the raw monthly data; tasks 4-5 (lab 5) move to yearly time series and small-multiple panels. Each figure below is annotated with what it shows and why the next step improves on it.

Task 2a - scatter with default markers



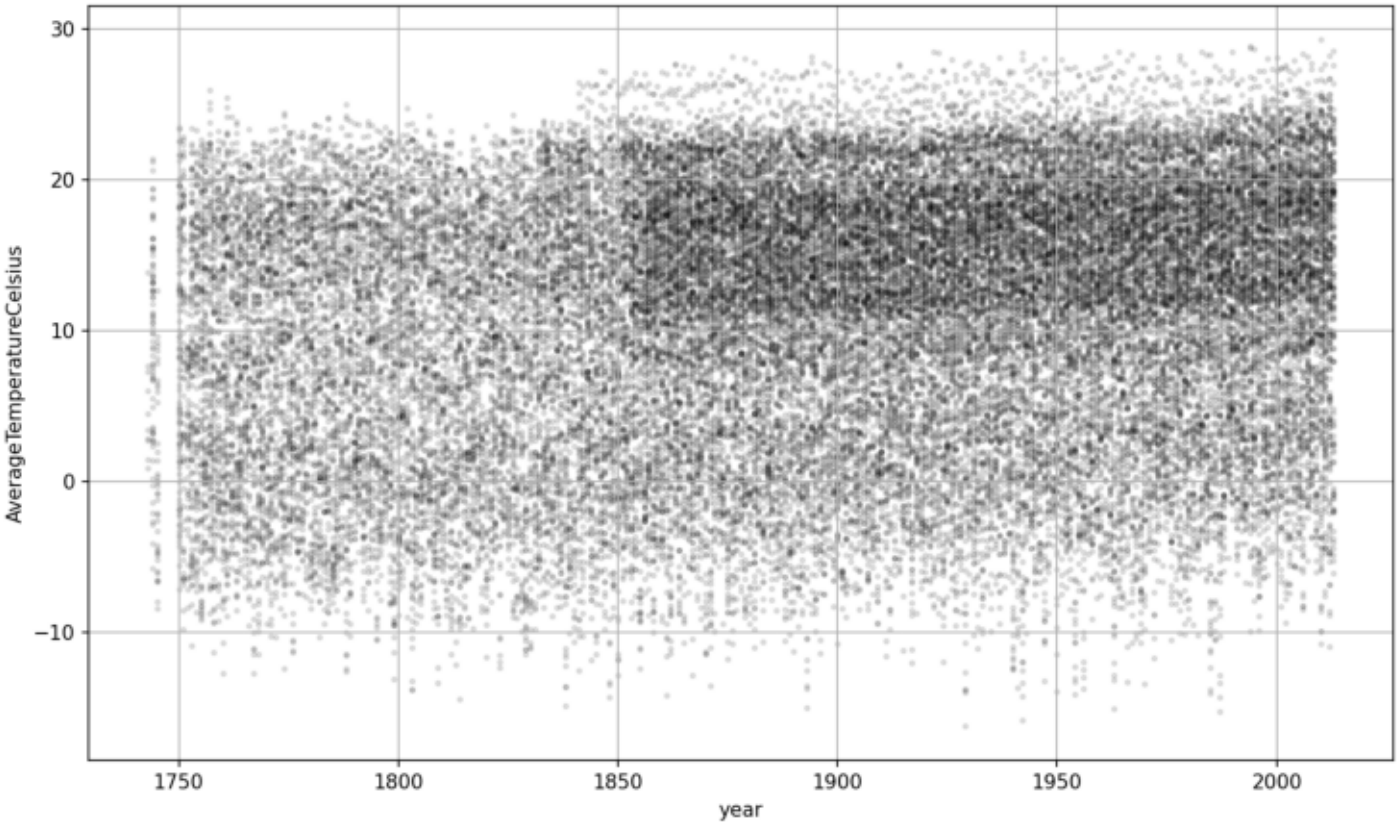
Every monthly average temperature against year. Large filled circles overlap so heavily that the plot is essentially a solid block: unreadable.

Task 2b - point markers and a grid



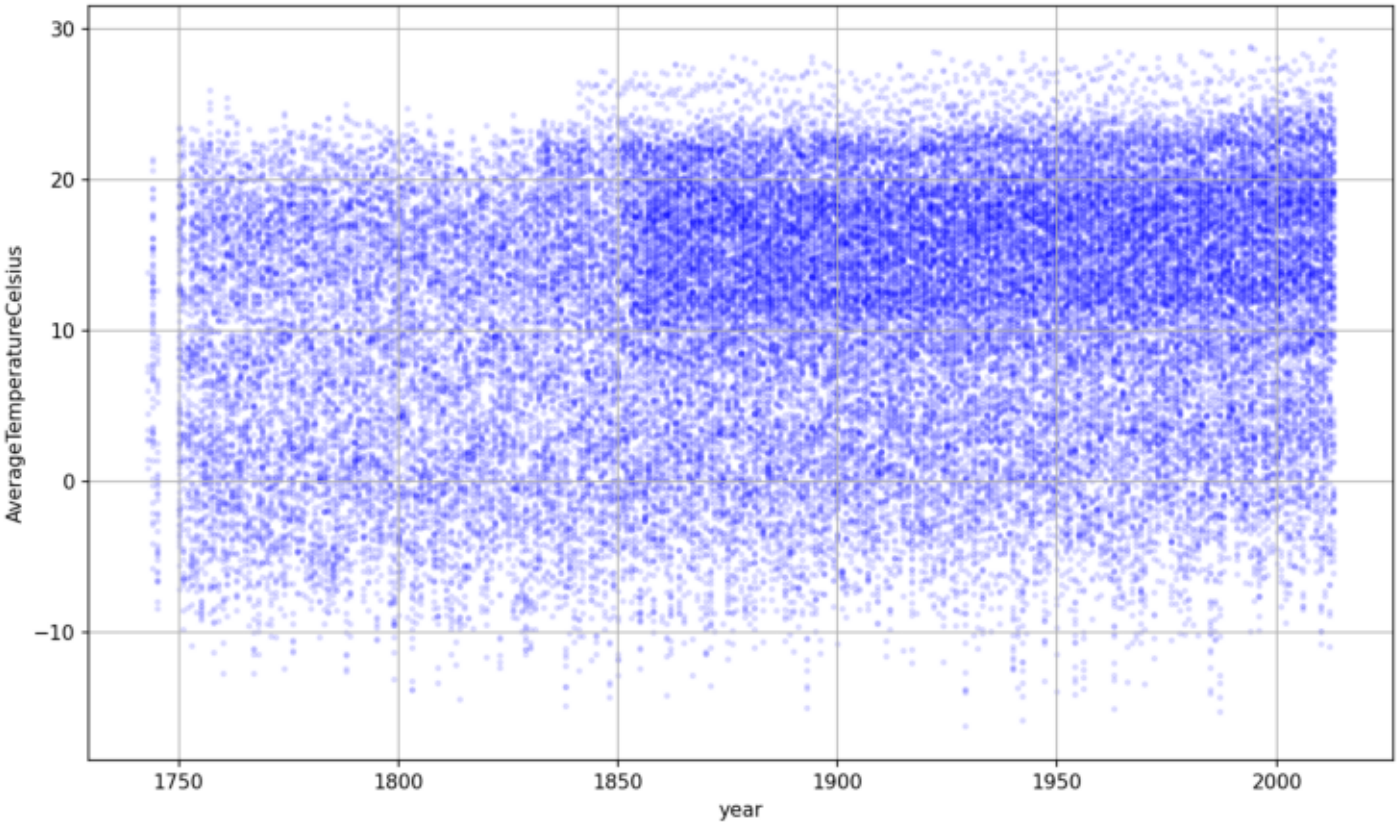
Switching to small point markers and adding a grid already reveals the vertical banding of the seasonal cycle, but the mass of points is still dense.

Task 2c - transparency



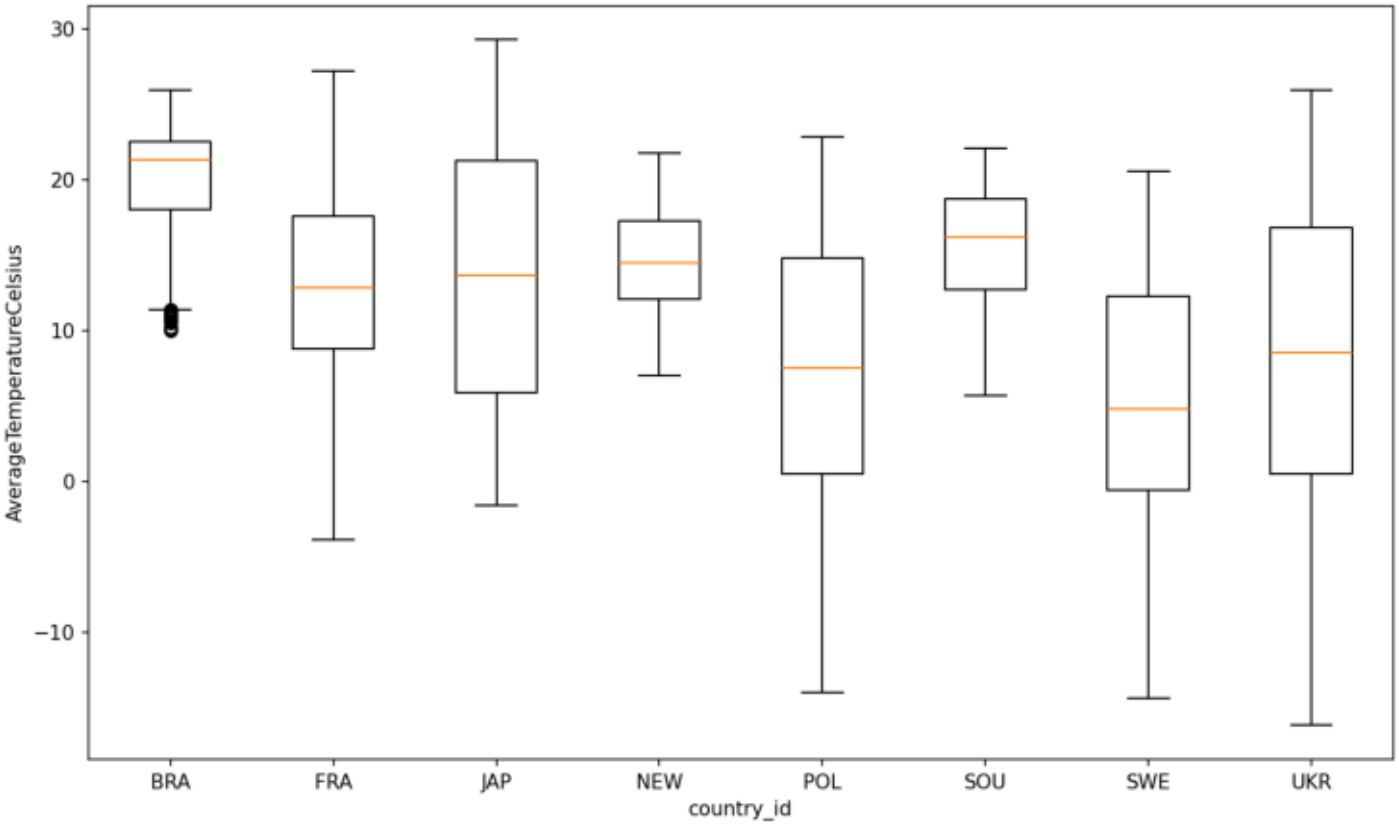
Adding alpha transparency lets density show through: darker regions are where most observations fall, so the bulk of the distribution becomes visible.

Task 2d - colour



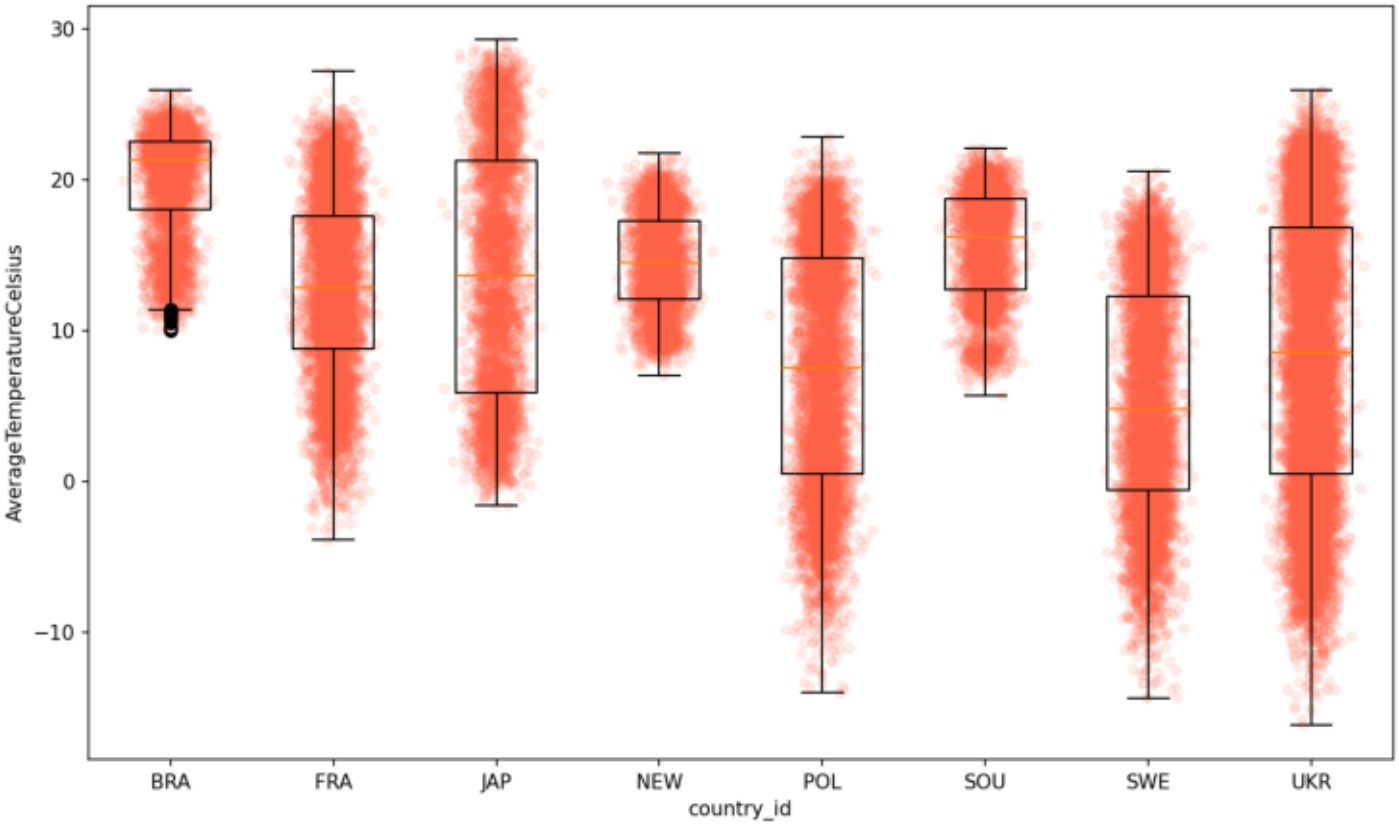
Colouring the points adds a third channel, but for this data it does not add much insight - a scatter plot is simply not the right tool here.

Task 3a - box plot per country



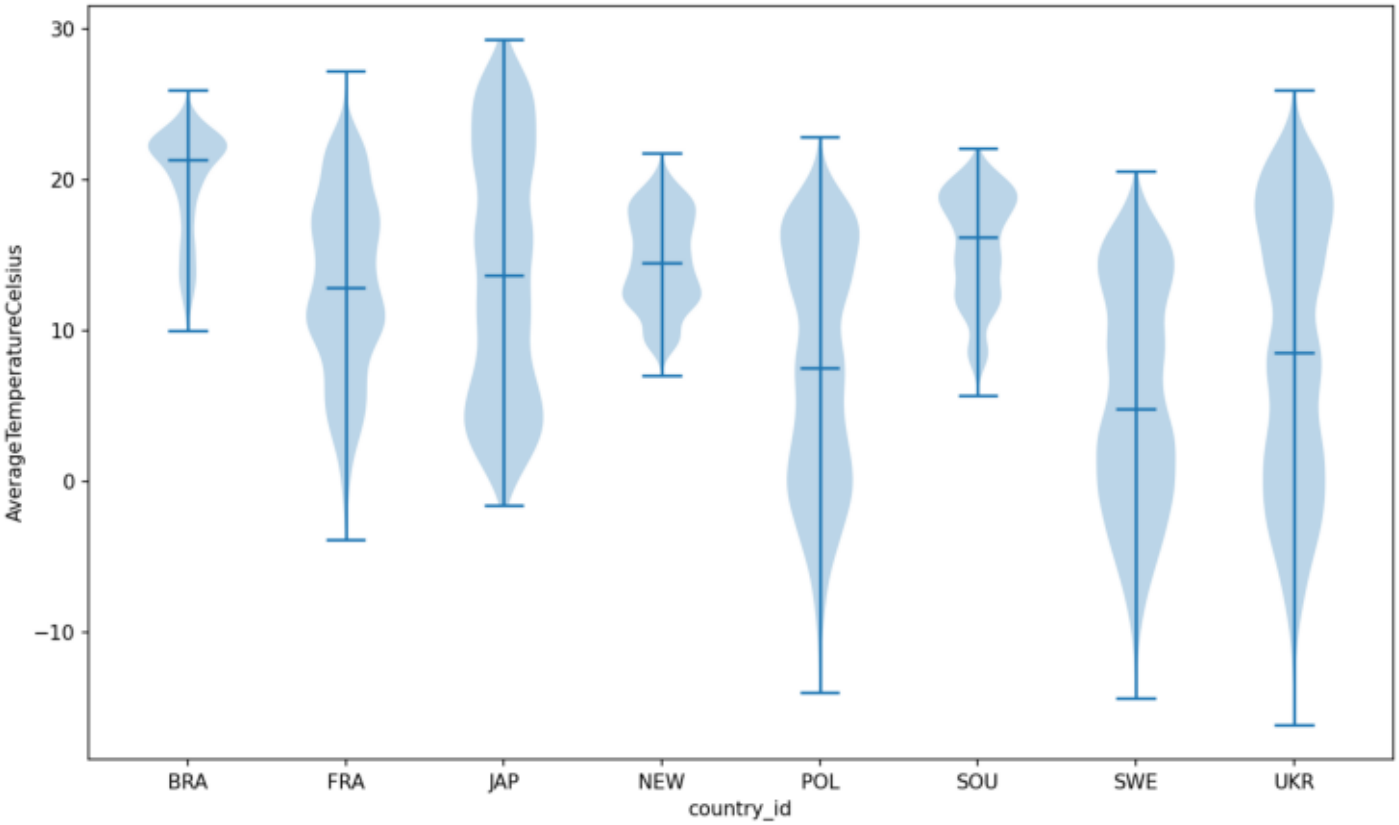
Distribution of temperatures within each country. Medians, spread and outliers are now readable at a glance, and hot vs. cold / stable vs. variable climates stand out.

Task 3b - box plot with jitter



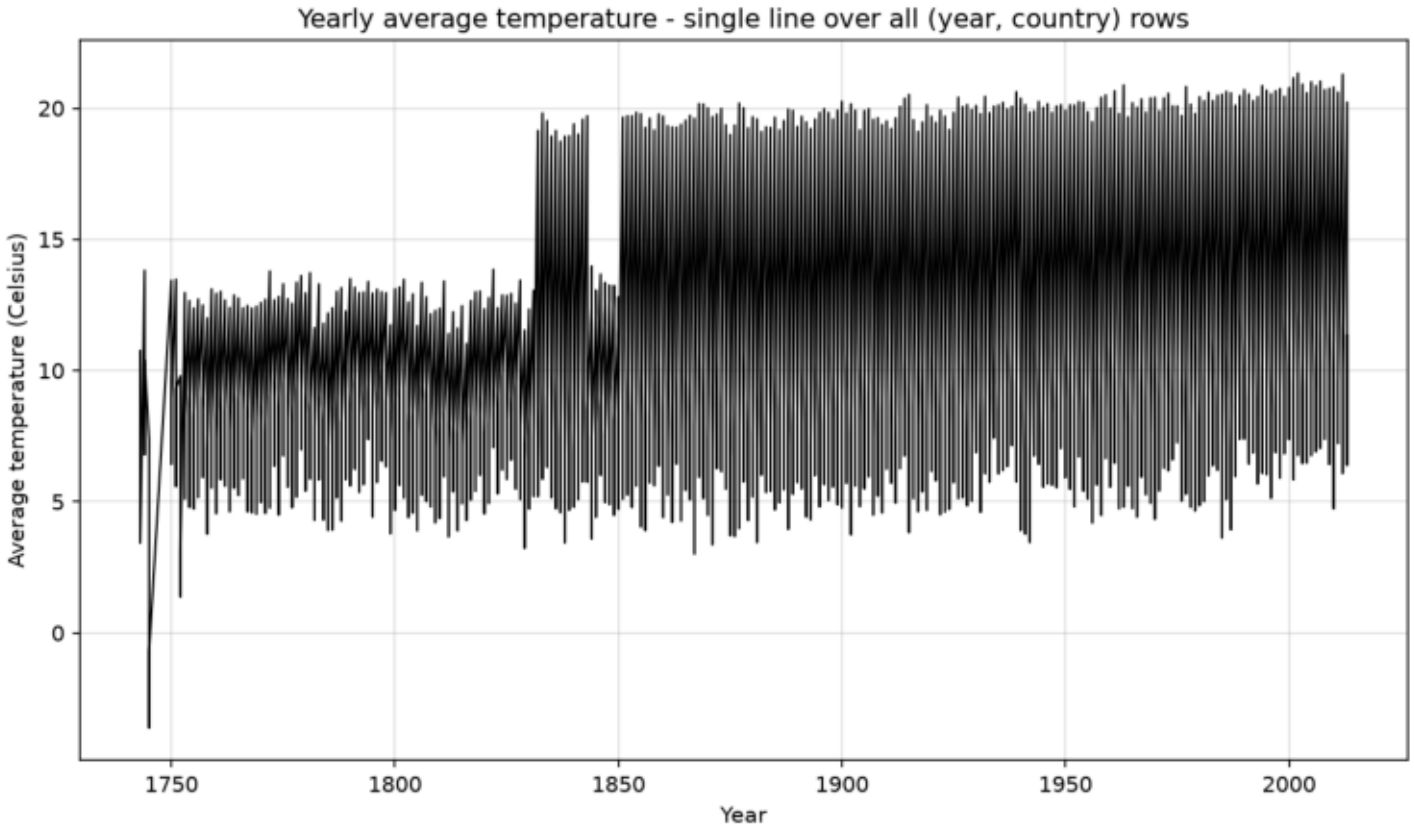
Overlaying a transparent jittered scatter shows how many observations sit behind each box, so sample size and density are not hidden by the summary.

Task 3c - violin plot



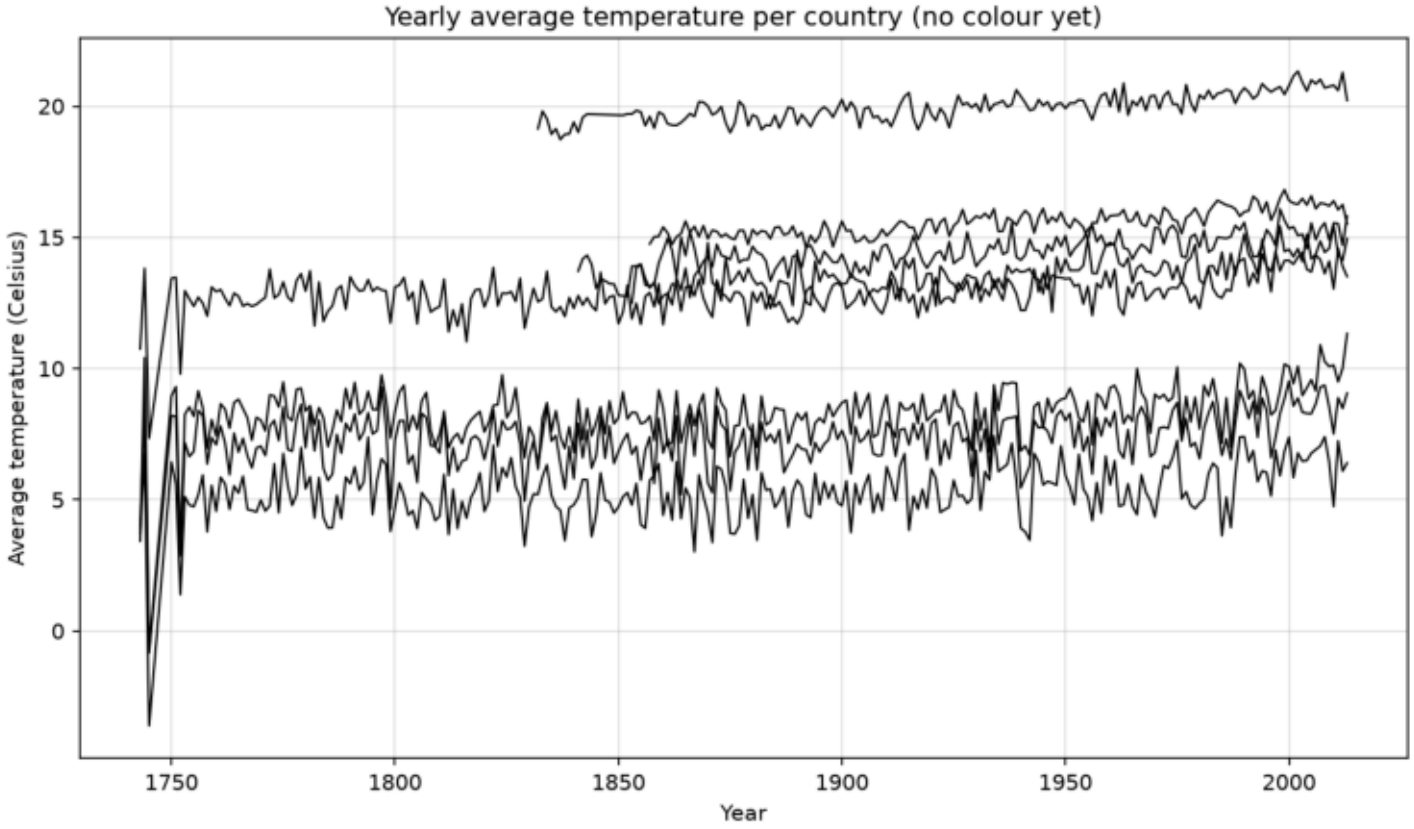
Violin plots draw the estimated density, exposing multi-modal shapes (e.g. strong seasonal double peaks) that a box plot would hide.

Task 4a - one continuous line



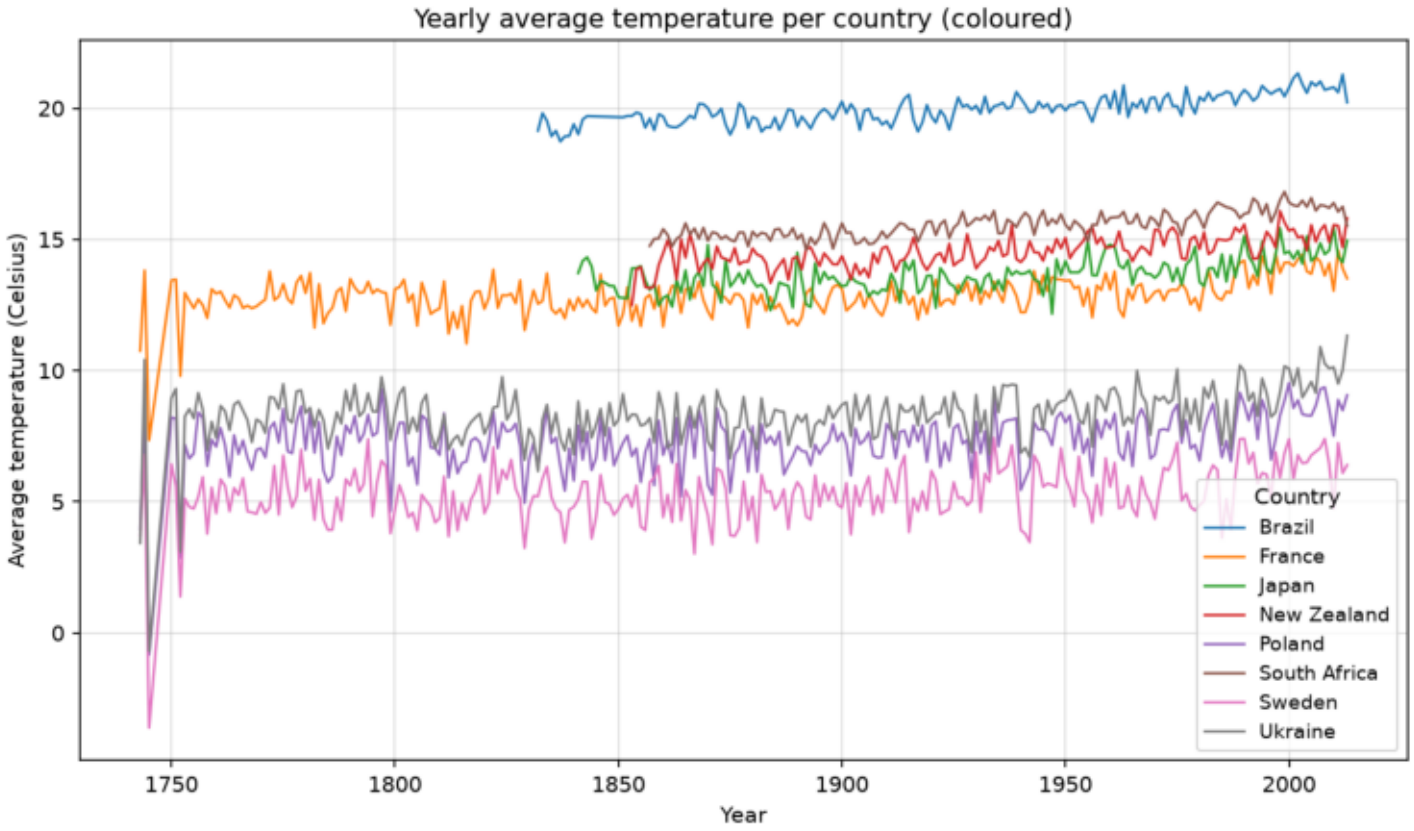
Yearly average per (year, country) plotted as a single line. The curve jumps between countries within each year, producing a jagged, unreadable trace.

Task 4b - split by country



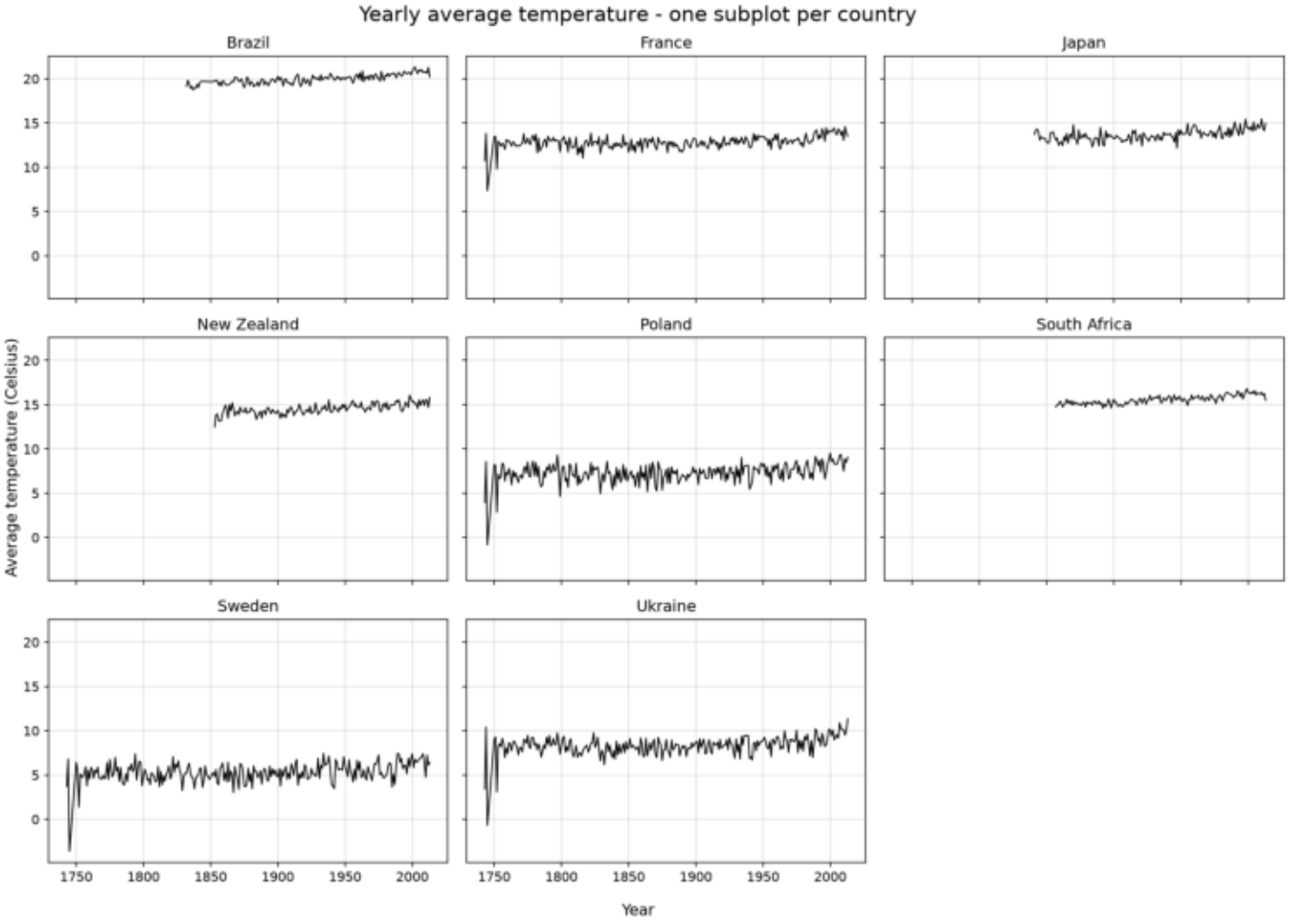
Drawing one black line per country removes the jumps; the individual climate levels separate, but without colour the lines are hard to tell apart.

Task 4c - colour per country



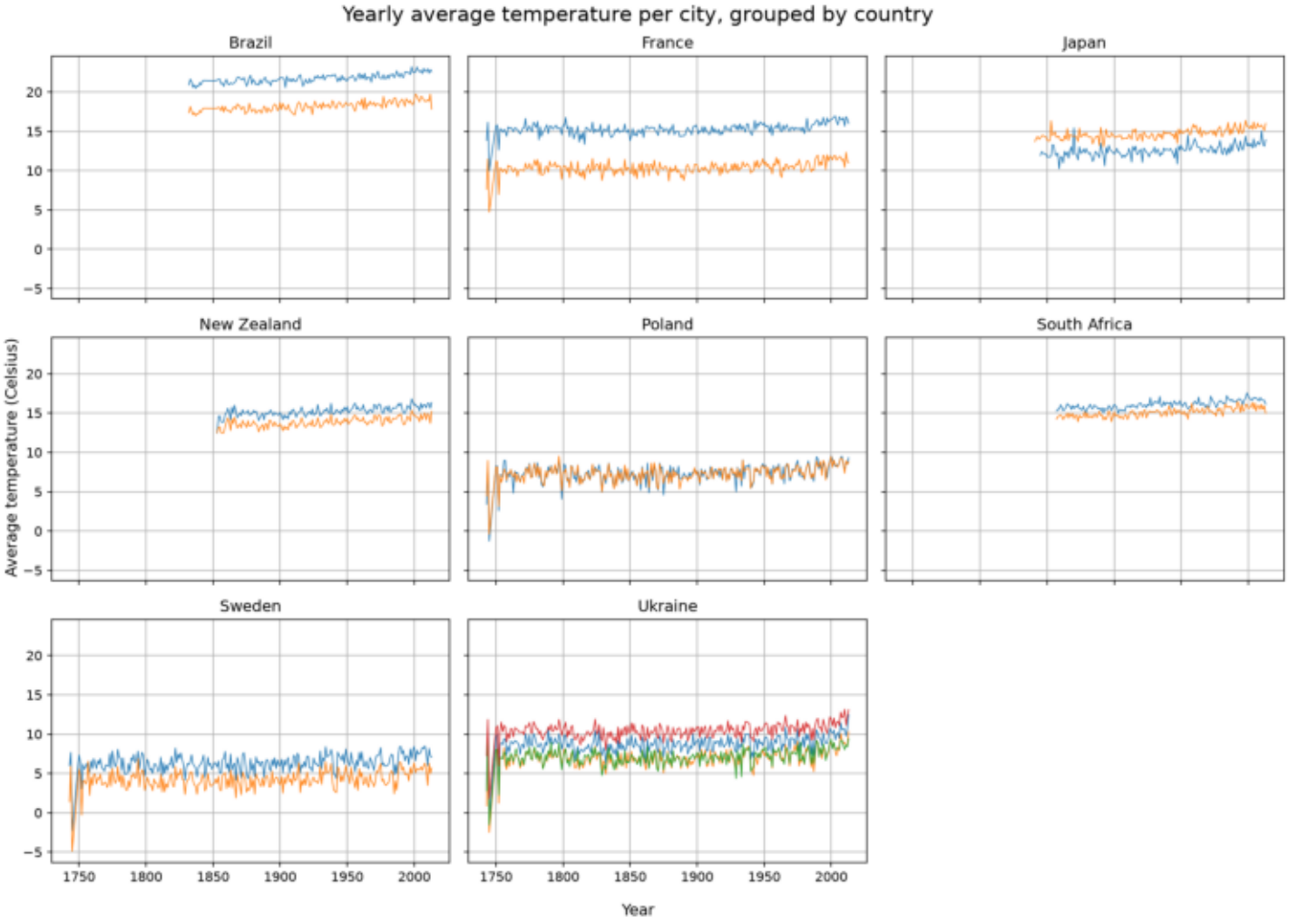
Assigning a colour and a legend entry to each country finally makes the yearly trends comparable; a gentle warming drift is visible in most series.

Task 5a - one subplot per country



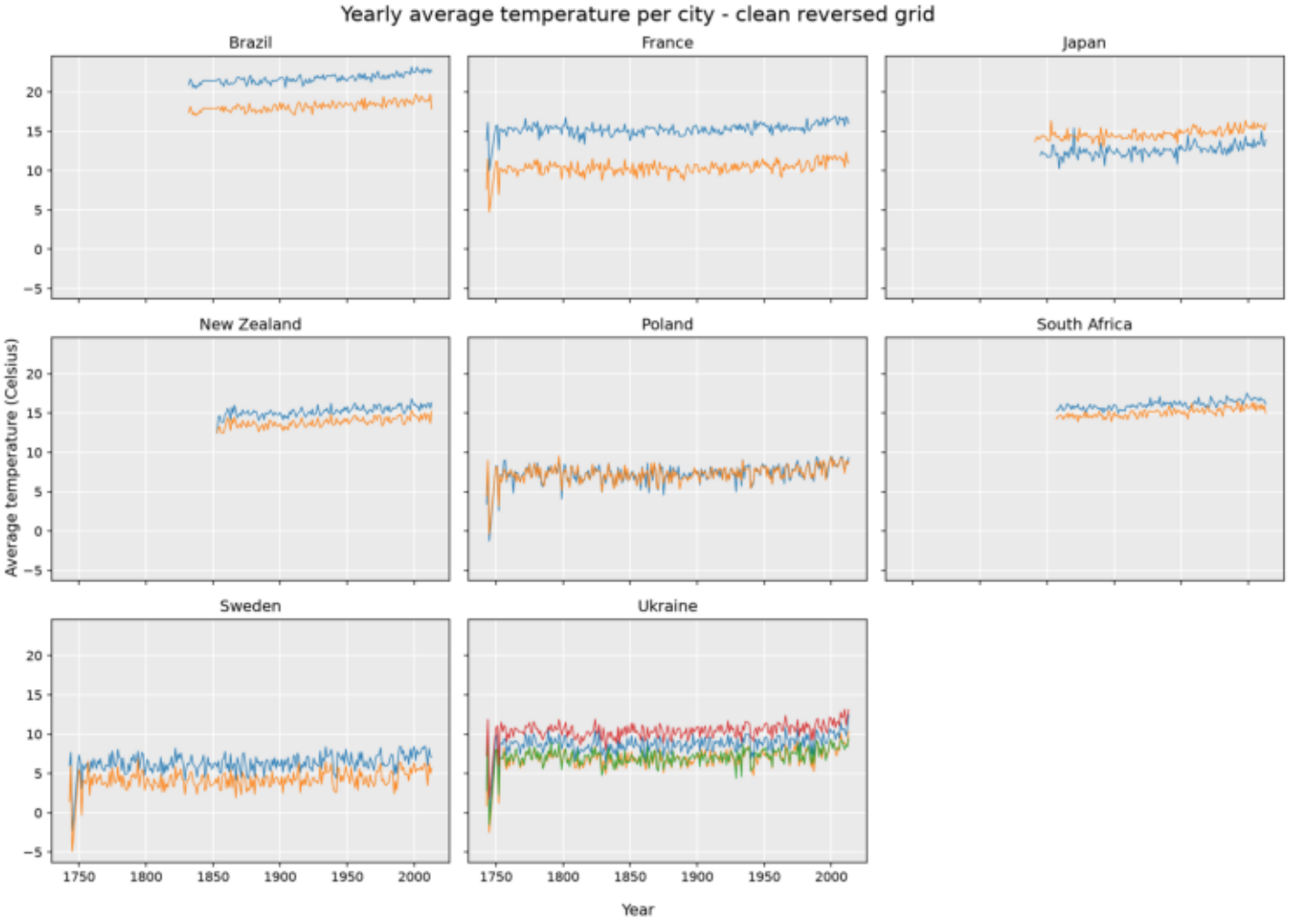
Faceting into a 3x3 panel gives each country its own axes, removing overlap and letting shape and trend be read country by country on a shared scale.

Task 5b - split each subplot by city



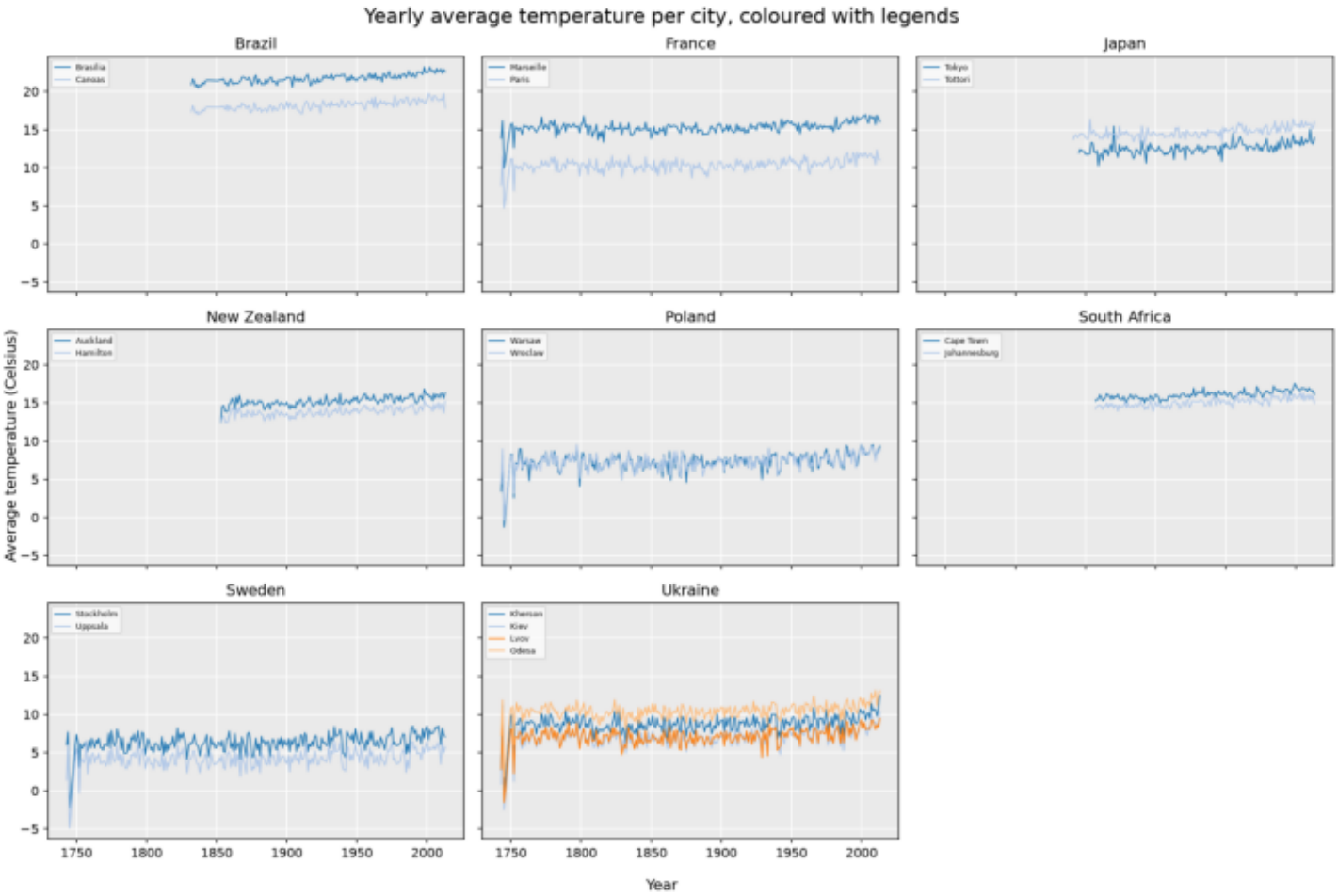
Each country panel is split into its cities. More detail is visible, but the default dark grid competes with the data.

Task 5c - clean reversed grid



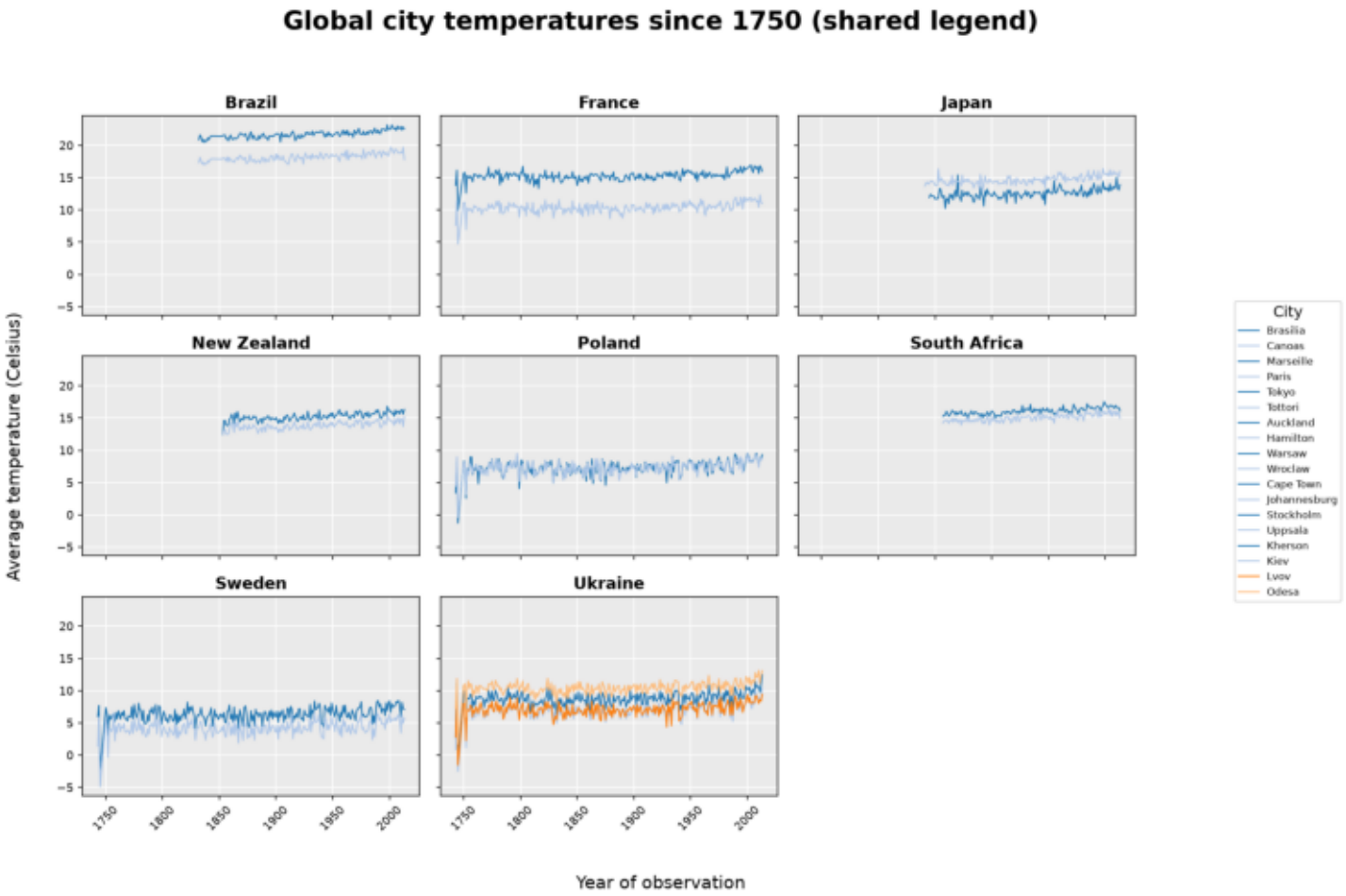
Reversing the grid (light panel, white grid lines) reduces visual noise and prints cleanly, while keeping the per-city structure.

Task 5d - cities in colour with legends



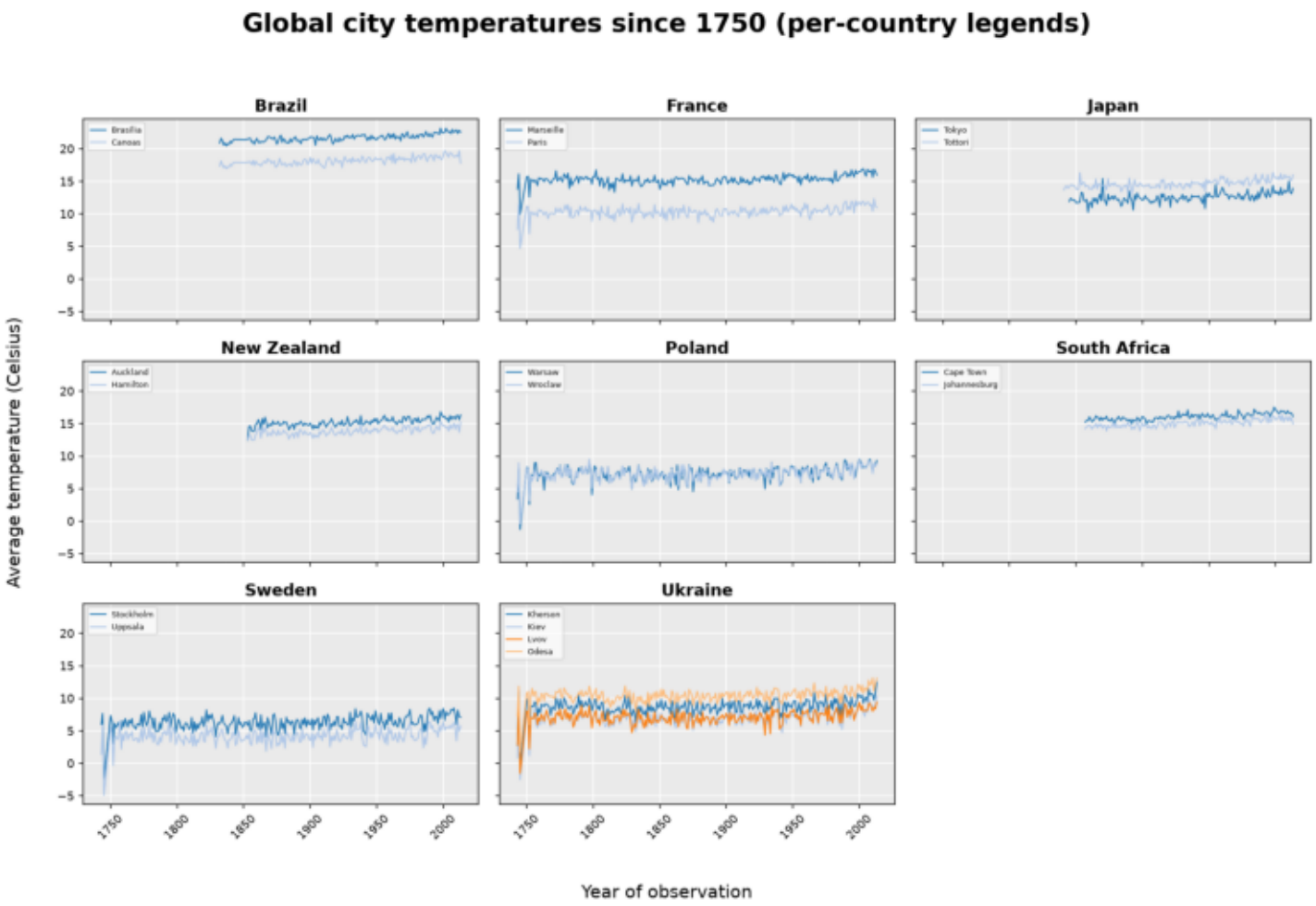
Colouring the cities and adding per-subplot legends uses the space gained by faceting to show which city each line is.

Task 5e - final panel (shared legend)



Presentation-ready version: bold titles, larger fonts, rotated year ticks and a single shared city legend on the right.

Task 5e - final panel (per-country legends)



Alternative final layout with a legend inside every country panel, useful when the figure is printed and read one panel at a time.